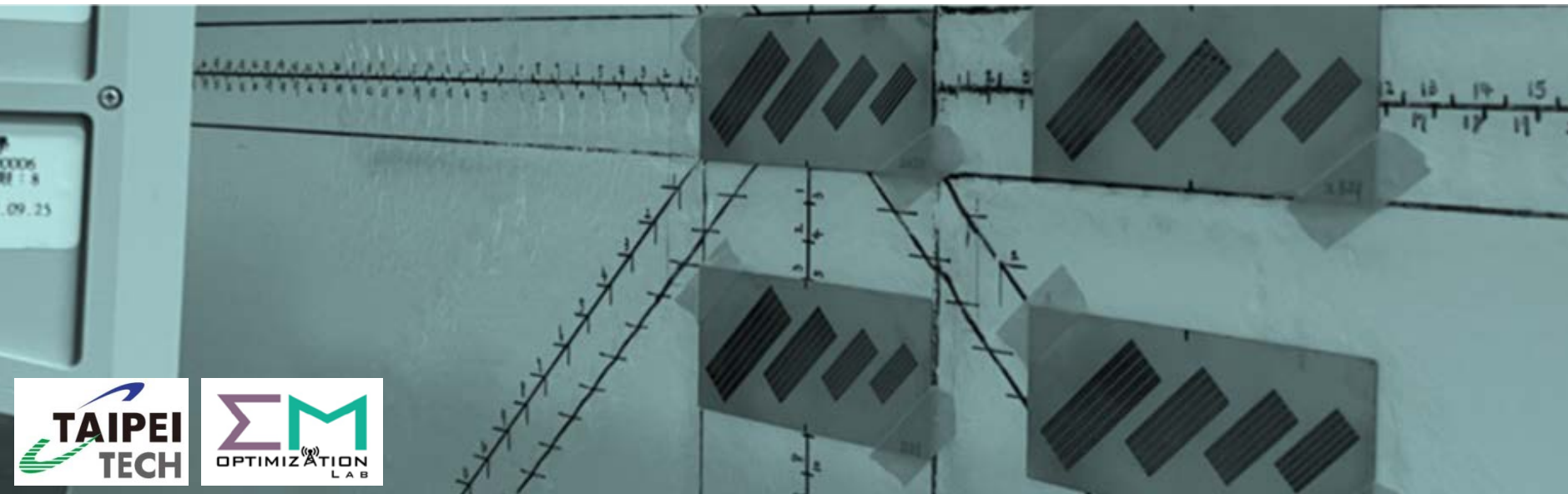
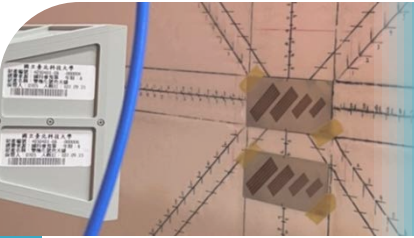


Linear Algebra

線性代數



Chapter 5: Eigenvalues and Eigenvectors
Yen-Sheng Chen (陳晏笙)
OCW Course Content



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Layer 2: Diagonalization

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Road to Spectral Theorem

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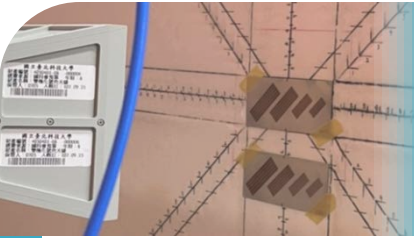
Layer 3: Spectral Theorem

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Layer 4: Positive Definite Matrices



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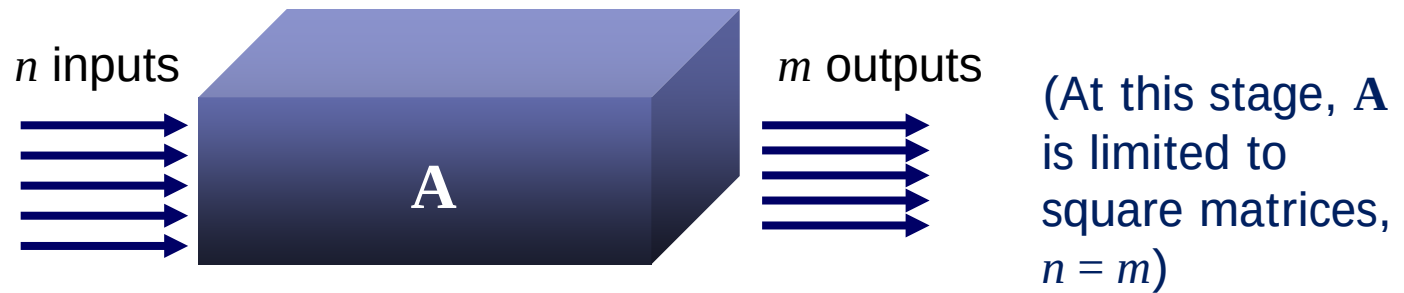
Layer 1: Jordan Form

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Layer 4: Positive Definite Matrices

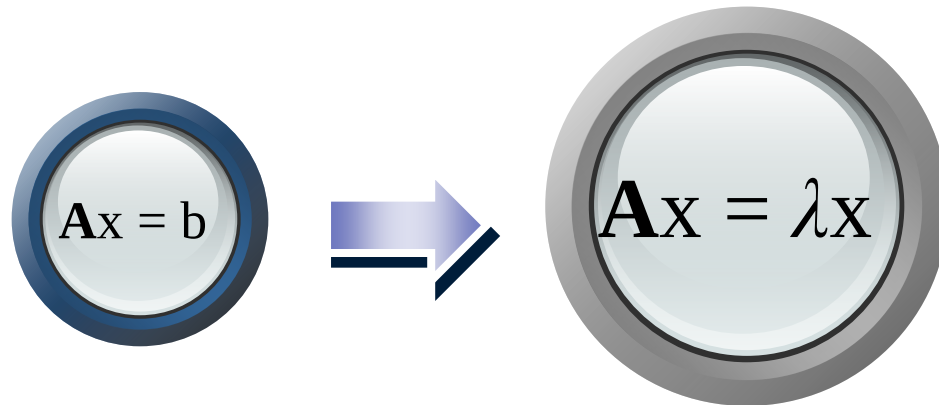
Recall Linear Transformations

- Recall the goal of this course:



- A is a process of “transformation”
- What transformations did we see in previous lectures?
 - ◆ Row operation, Gaussian elimination, and permutation
 - ◆ Stretch, rotation, and reflection
 - ◆ Projection and Gram-Schmidt
- Are these transformations computationally efficient?
- What if we deal with higher-dimensional inputs?

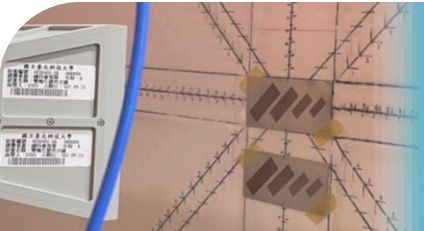
Simplify the Transformation!



The transformation by
a matrix $\mathbf{A}$

The transformation by a scalar λ ,
that is, a multiple of the vector $\mathbf{x}$

- The computational effort is greatly reduced
- We simplify the procedure drastically
- But, can we do this for any inputs ($\mathbf{x}$)?
- If not, what inputs lead $\mathbf{Ax}$ to $\lambda \mathbf{x}$?

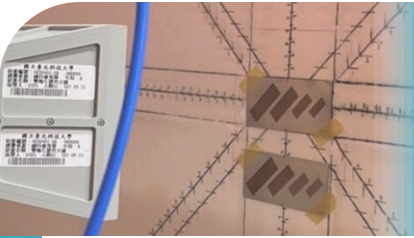


Road to Eigenvalues

- Only some inputs $\mathbf{x}$ lead to $\mathbf{Ax} = \lambda\mathbf{x}$, so these $\mathbf{x}$ are unknowns
- λ is the key to this simplification, so λ is also unknown
- This makes $\mathbf{Ax} = \lambda\mathbf{x}$, or, $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = 0$, a nonlinear problem
- However, if we get λ first and solve the associated $\mathbf{x}$ later, $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = 0$ becomes a linear problem

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = 0$$

- λ is chosen so that $\mathbf{A} - \lambda\mathbf{I}$ has a nullspace $\rightarrow \det(\mathbf{A} - \lambda\mathbf{I}) = 0$
- The vector $\mathbf{x}$ is in the nullspace of $(\mathbf{A} - \lambda\mathbf{I})$
- $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$ is called the “characteristic equation”

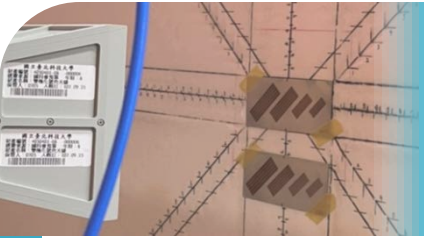


Solving the Characteristic Equation

$$(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = 0, \text{ or } \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

- Each solution λ has a corresponding vector $\mathbf{x}$
- λ is called the eigenvalue of the transformation matrix $\mathbf{A}$
- $\mathbf{x}$ is called the eigenvector of the corresponding λ
- Example: Find λ and $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$

$$\mathbf{A} = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix}$$

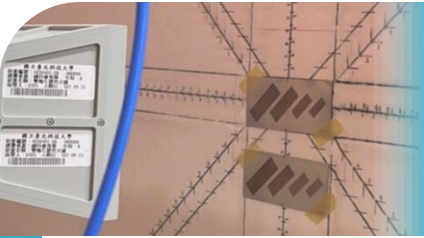


Why Eigenvectors Are Useful?

- The simplification of $\mathbf{A}\mathbf{x}$ to $\lambda\mathbf{x}$ is for only “certain” inputs
- Usually, the interested input will not be as parallel to the eigenvectors
- Then, why do we concern such simplification?

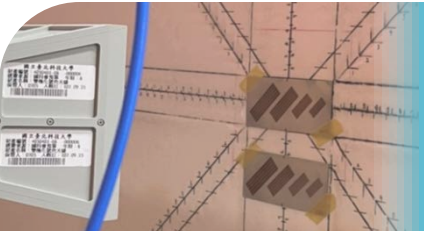
Recall the Importance of a Basis...

- If you can find n independent eigenvectors that span R^n , what can you do for all the vectors in this space?
- These eigenvectors form a basis, and the space spanned by these basis vectors is called “eigenspace”



Concept clarification

- For an $n \times n$ matrix $\mathbf{A}$:
 - ◆ The number of eigenvalues:
 - ◆ The number of nonzero eigenvalues:
 - ◆ The number of (repeated) zero eigenvalues:
 - ◆ Can nonzero eigenvalues repeat?
 - ◆ What happens when eigenvalues repeat?
 - ◆ What if all eigenvalues are distinct?

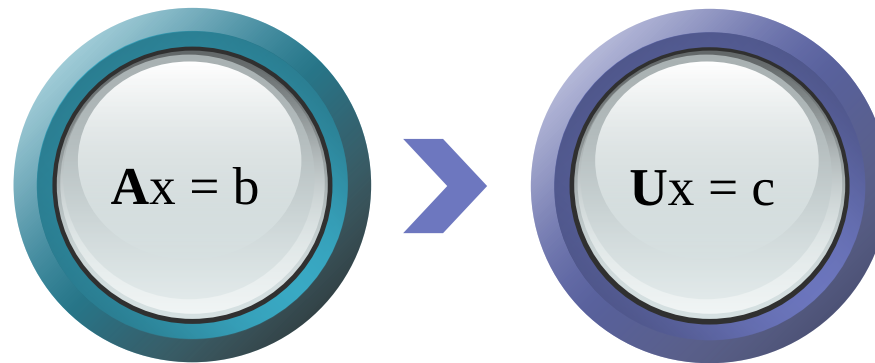


A New Point of View of Singularity

- If a matrix is singular, it is non-invertible
- Previously, singularity of a squared matrix $\mathbf{A}$ is observed from
 - ◆ $\det \mathbf{A} = 0$
 - ◆ The number of pivots $r < n$
 - ◆ $\mathbf{A}$ has a nontrivial nullspace
 - ◆ $\mathbf{A}$ has a nontrivial left nullspace
 - ◆ Linearly dependent columns
 - ◆ Linearly dependent rows
 - ◆ Dimension of $C(\mathbf{A}) < R^n$
 - ◆ Dimension of $C(\mathbf{A}^T) < R^n$
- A new tool for identifying singular matrix $\mathbf{A}$: $\lambda = 0$ exist(s)
- For a matrix with $\lambda = 0$: $\mathbf{A}$ has a non-trivial nullspace
- What is the eigenvector for $\lambda = 0$?

- Example: projection matrix $\mathbf{P}_1 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ $\mathbf{P}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

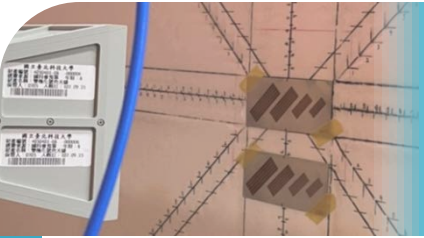
How Do We Find Eigenvalues?



- When we attempted to find the four subspaces of $\mathbf{A}$, we used to do the Gaussian elimination to get $\mathbf{U}$
- It is easy to find the eigenvalues from $\mathbf{U}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 4 & 8 \end{bmatrix} \Rightarrow \mathbf{U} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 4 & 4 \\ 0 & 0 & 2 \end{bmatrix} \Rightarrow \begin{vmatrix} 1-\lambda & 0 & 1 \\ 0 & 4-\lambda & 4 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$$

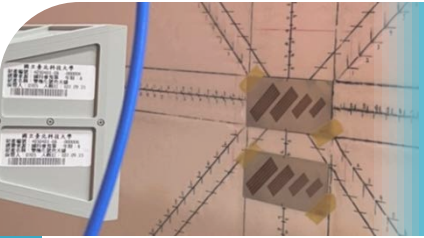
- However, the eigenvalues in $\mathbf{A}$ and those in $\mathbf{U}$ are different!



How Do We Find Eigenvalues?

- Gaussian elimination changes the eigenvalues
- Numerical methods are required to find the eigenvalues
- Some checks on the eigenvalues:

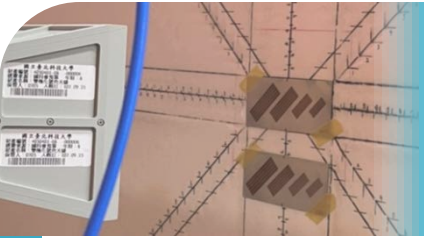
1. $\lambda_1 + \lambda_2 + \dots + \lambda_n = a_{11} + a_{22} + \dots + a_{nn} = \text{trace}$
2. $\lambda_1 \times \lambda_2 \times \dots \times \lambda_n = \det \mathbf{A}$



Special Cases

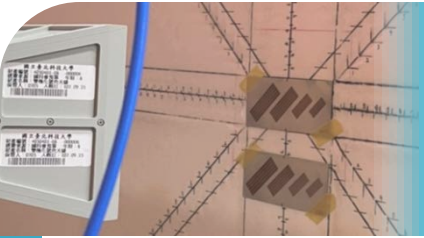
- If $\mathbf{A}$ is invertible with eigenvalues of λ , the eigenvalue of $\mathbf{A}^{-1}$ is:
- The eigenvectors of $\mathbf{A}^{-1}$ are:

- If $\mathbf{A}$ has eigenvalues of λ , the eigenvalue of $\mathbf{A}^T$ is:
- The eigenvectors of $\mathbf{A}^T$ are:



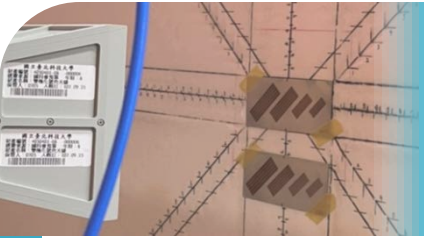
ConceptTests

- How many eigenvalues can a 5×3 matrix offer?
 - 1) 5
 - 2) 3
 - 3) It depends. We know the number only when we have solved the characteristic equation
 - 4) Nonsense. 0



ConceptTests

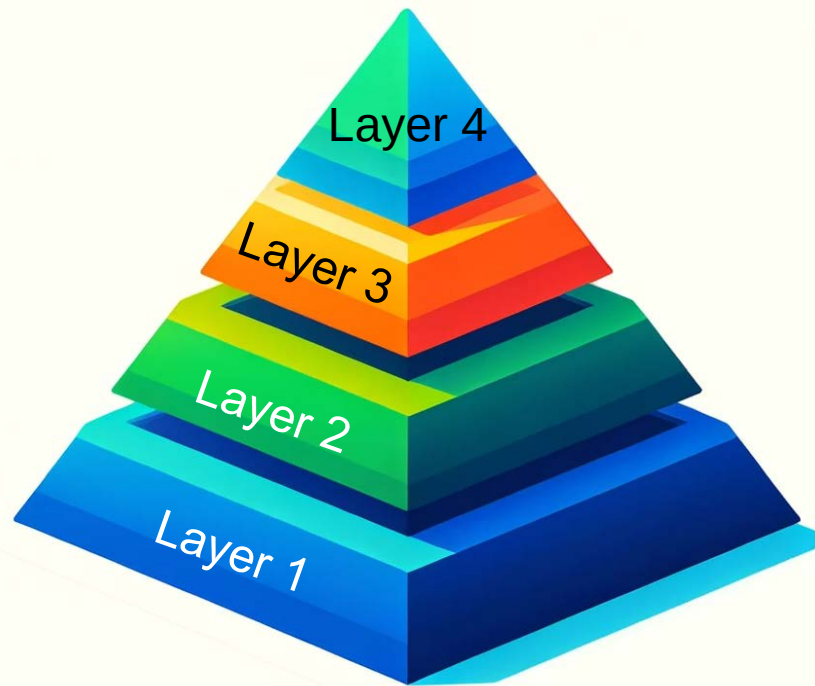
- What does it mean if zero is an eigenvalue of a matrix $\mathbf{A}$?
 - 1) The matrix $\mathbf{A}$ is invertible
 - 2) The matrix $\mathbf{A}$ is singular
 - 3) The determinant of $\mathbf{A}$ is zero
 - 4) The matrix $\mathbf{A}$ cannot provide a full set of eigenvectors
 - 5) The matrix $\mathbf{A}$ has a non-trivial nullspace



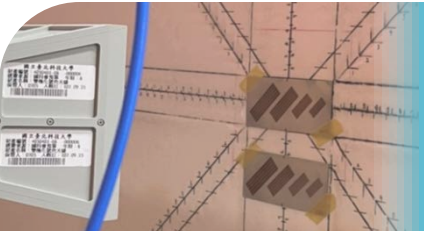
ConceptTests

- Why do we find eigenvalues and eigenvectors of a matrix?
 - 1) To determine the rank of the matrix
 - 2) To simplify the process of the transformation
 - 3) To compute the determinant from the multiplication of eigenvalues
 - 4) To find the inverse of a matrix

Roadmap in This Chapter



Rectangular matrices



Examples of Defective Matrix

- 2×2 :

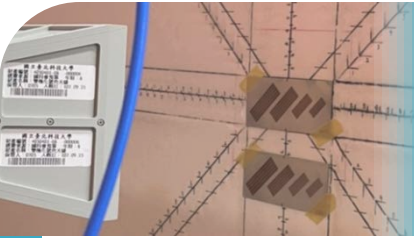
$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

- Express λ in terms of a , b , c , and d :
- Sufficient condition for having a full set of independent eigenvectors:
- Examples for defective matrix:

- 4×4 :

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- Can $\mathbf{A}$ be diagonalized?
- If not, is there any way to fix $\mathbf{A}$?



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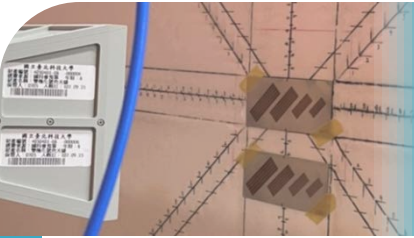
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Diagonal Form of a Matrix

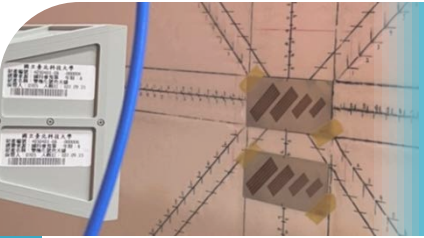
$$\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1} \quad \text{or} \quad \mathbf{\Lambda} = \mathbf{X}^{-1}\mathbf{A}\mathbf{X}$$

- $\mathbf{\Lambda}$: Diagonal matrix formed by eigenvalues
- $\mathbf{X}$: corresponding eigenvectors on the columns

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$$

Requirements for diagonalization

- Squared $\mathbf{A}$ ($n \times n$)
- Independent eigenvectors
 - ◆ If all eigenvalues are distinct, all eigenvectors are independent
→ the matrix can be diagonalized
 - ◆ However, for repeated eigenvalues, it is still possible to find independent eigenvectors



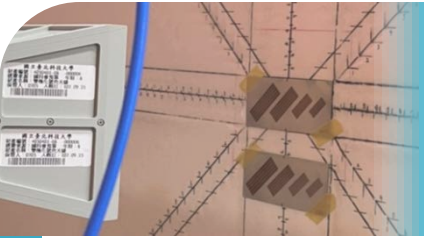
Diagonal Form of a Matrix

$$\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1} \quad \text{or} \quad \mathbf{\Lambda} = \mathbf{X}^{-1}\mathbf{A}\mathbf{X}$$

- The factorization is not unique
- Not all matrices are diagonalizable
- Examples:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$$

- Diagonalizability is concerned with the eigenvectors
- Invertibility is concerned with the eigenvalues
- If the eigenvectors correspond to distinct eigenvalues then those eigenvectors are linearly independent



Examples for Diagonalization

- Projection matrix:

$$\mathbf{A} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

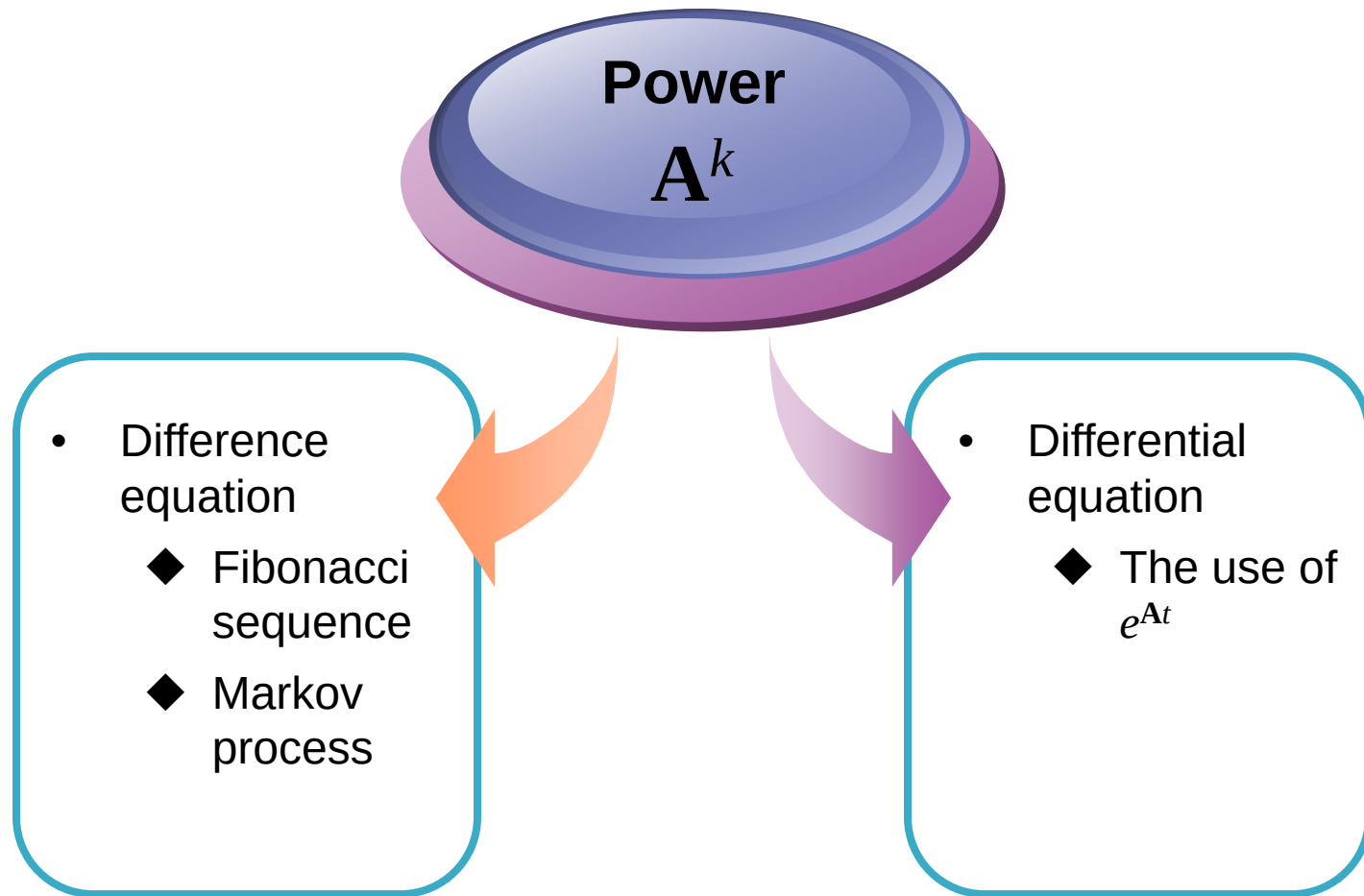
- Rotation matrix (skew symmetric matrix):

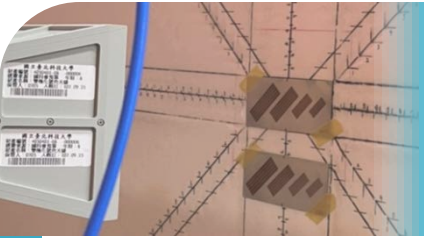
$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{bmatrix}$$

- Symmetric matrix:

$$\mathbf{A} = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

Applications of Diagonalization



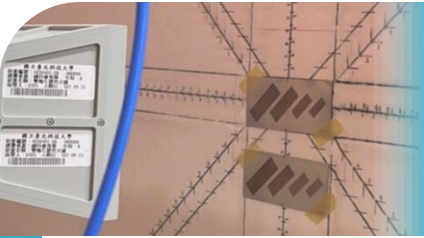


Powers A^k

- An application of diagonalization: the power of a matrix
- $A = X\Lambda X^{-1} \rightarrow A^k =$
- Example: Rotation matrix through 90°

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$A^4 =$$



An Example

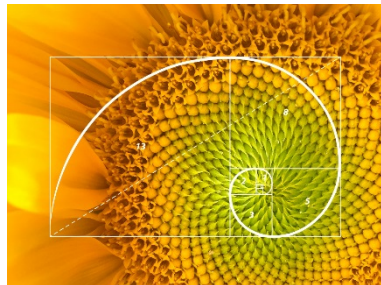
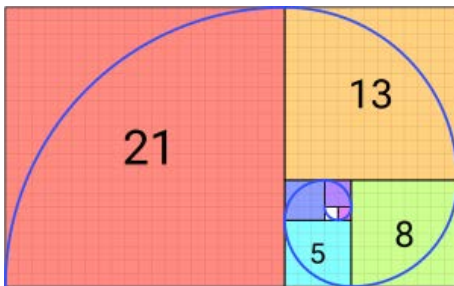
$$\mathbf{A} = \begin{bmatrix} -1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$

- Find the three eigenvalues and the corresponding eigenvectors.
- Compute $\mathbf{A}^{1001}$.

Difference Equation

- Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13,...

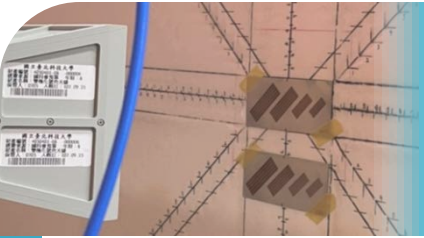
$$F_{k+2} = F_{k+1} + F_k$$



Source: 1. https://en.wikipedia.org/wiki/Fibonacci_sequence

2. Jonathan Cleveland, The nature of design: the Fibonacci sequence and the Golden Ratio, Expertise, Sept. 2020

- Question: 1000th Fibonacci number?



Difference Equation

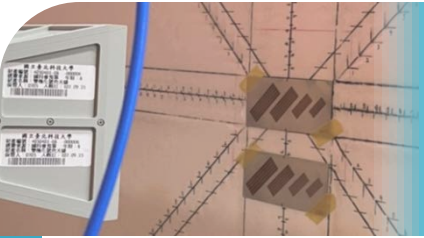
- Reduce the problem into $u_{k+1} = \mathbf{A}u_k$:

$$u_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$

$$\Rightarrow u_{k+1} = \begin{bmatrix} \quad \\ \quad \end{bmatrix} u_k$$

- The second equation is trivial and is a standard trick for second order difference equation
- For higher order equations, say order s , we need $s - 1$ trivial equations

$$\Rightarrow u_{k+1} = \mathbf{A}u_k \rightarrow u_k = \mathbf{A}^k u_0$$



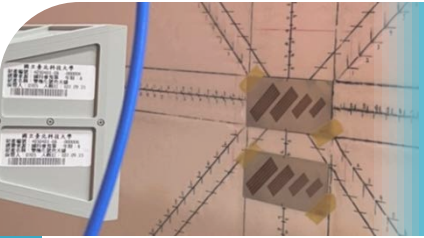
Difference Equation

$$u_k = A^k u_0 = (X \Lambda X^{-1})(X \Lambda X^{-1}) \cdots (X \Lambda X^{-1}) u_0 = X \Lambda^k \underbrace{X^{-1} u_0}_{\text{Set as } c}$$

- By setting u_0 as the linear combination of the eigenvectors, the solution becomes

$$u_k = X \Lambda^k c = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} \lambda_1^k \\ \lambda_2^k \\ \ddots \\ \lambda_n^k \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = c_1 \lambda_1^k x_1 + \cdots + c_n \lambda_n^k x_n$$

- The vector c is determined by the initial conditions



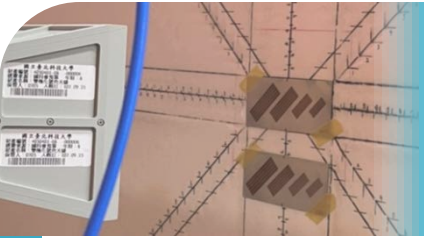
Difference Equation

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \rightarrow \lambda_1 = \quad , \lambda_2 =$$

- And initial conditions: $\mathbf{c} = \mathbf{X}^{-1}\mathbf{u}_0 = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\lambda_1 - \lambda_2} \\ \frac{-1}{\lambda_1 - \lambda_2} \end{bmatrix}$
- We have $\mathbf{u}_k = c_1 \lambda_1^k \mathbf{x}_1 + c_2 \lambda_2^k \mathbf{x}_2$

$$\Rightarrow F_k = \frac{\lambda_1^k}{\lambda_1 - \lambda_2} - \frac{\lambda_2^k}{\lambda_1 - \lambda_2} = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^k - \left(\frac{1 - \sqrt{5}}{2} \right)^k \right]$$

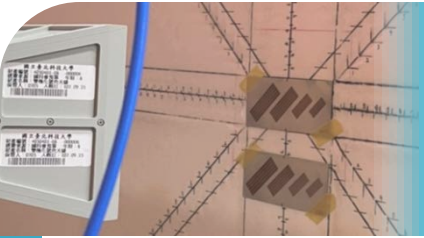
$$\Rightarrow \frac{F_{k+1}}{F_k} \approx \frac{1 + \sqrt{5}}{2} \approx 1.618 \quad \textit{Golden ratio!}$$



Similar Examples

If each number is the average of the two previous numbers,
 $G_{k+2} = \frac{1}{2}(G_{k+1} + G_k)$, set up the matrix $\mathbf{A}$ to update the vector
 $\begin{bmatrix} G_{k+2} \\ G_{k+1} \end{bmatrix} = \mathbf{A} \begin{bmatrix} G_{k+1} \\ G_k \end{bmatrix}$. We already know that $G_0 = 0$ and $G_1 = 1/2$.

- Find the eigenvalues and eigenvectors for $\mathbf{A}$.
- Find the limit as $k \rightarrow \infty$ for the matrix $\mathbf{A}^k$.
- Compute G_k as $k \rightarrow \infty$.

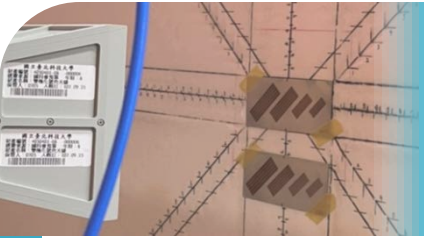


Markov Process

Markov matrix

- Each column adds up to 1
 - All entries are nonnegative
 - There is always an eigenvalue $\lambda_1 = 1$
 - Other eigenvalues $|\lambda_i| \leq 1$
-
- Each year 1/10 of the people outside Taipei move in, and 2/10 of the people inside Taipei move out. What is the population of Taipei after 5 years, 10 years, or 100 years?
 - Population outside: y . Inside: z

$$\begin{bmatrix} y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} y_0 \\ z_0 \end{bmatrix}$$



Example

- Solution:

$$\begin{bmatrix} y_k \\ z_k \end{bmatrix} = \mathbf{A}^k \begin{bmatrix} y_0 \\ z_0 \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} y_0 \\ z_0 \end{bmatrix}$$

- Steady-state solution:
- The steady state is the eigenvector of $\mathbf{A}$ corresponding to $\lambda_1 = 1$

System Stability

Stability	Example	Eigenvalue
Stable	Reduced to 0	All $ \lambda_i < 1$
Neutrally stable	Markov process	Some $ \lambda_i = 1$ and other $ \lambda_i < 1$
Unstable	Fibonacci number	At least one $ \lambda_i > 1$

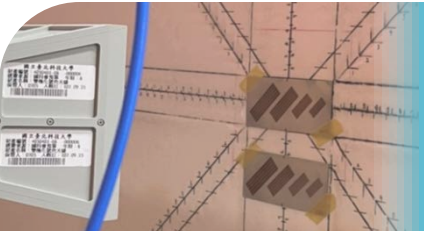
$$\mathbf{u}_k = \mathbf{X}\mathbf{\Lambda}^k\mathbf{X}^{-1}\mathbf{u}_0 = c_1\lambda_1^k\mathbf{x}_1 + \cdots + c_n\lambda_n^k\mathbf{x}_n$$

- The behavior of the system can be fully observed using λ_i and $\mathbf{x}_i$
- Eigenvalues and eigenvectors show us how this system works
- Example:

$$\mathbf{A} = \begin{bmatrix} 0 & 4 \\ 0 & \frac{1}{2} \end{bmatrix} \text{ and } \mathbf{u}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The first step is to split $\mathbf{u}_0$ into two eigenvectors:

$$\mathbf{u}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -8 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 8 \\ 1 \end{bmatrix}$$



Applying the Experience for Finding Eigenvalues

- Example 1:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

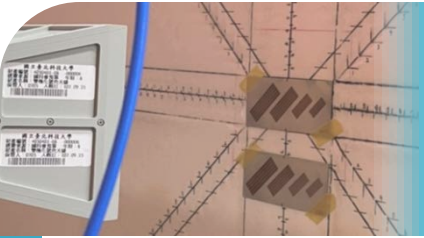
- Example 2:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

- Example 3:

$$\mathbf{A} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$$

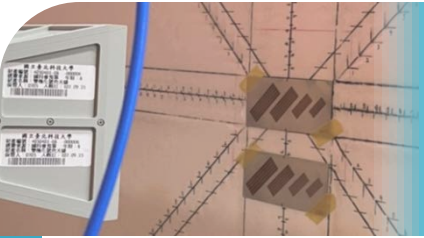
Find the pattern in eigenvalues?
Find the orthogonality in eigenvectors?



More Examples

$$\mathbf{A} = \begin{bmatrix} 110 & 55 & -164 \\ 42 & 21 & -62 \\ 88 & 44 & -131 \end{bmatrix}$$

- Find the three eigenvalues of $\mathbf{A}$
- Compute $\mathbf{A}^{2025}$



Exponential of a Matrix

- What is exponential of $\mathbf{A}$: $e^{\mathbf{A}t}$
- Recall Taylor series: $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

$$\Rightarrow e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{(\mathbf{A}t)^2}{2!} + \frac{(\mathbf{A}t)^3}{3!} + \dots$$

- Relate this to diagonalization

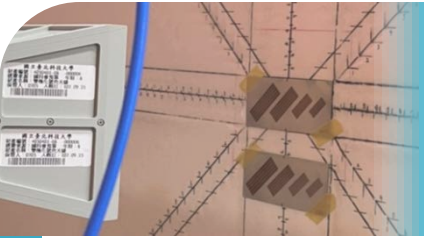
$$\begin{aligned} e^{\mathbf{A}t} &= \mathbf{I} + \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}t + \frac{\mathbf{X}\mathbf{\Lambda}^2\mathbf{X}^{-1}t^2}{2!} + \frac{\mathbf{X}\mathbf{\Lambda}^3\mathbf{X}^{-1}t^3}{3!} + \dots \\ &= \mathbf{X} \left(\mathbf{I} + \mathbf{\Lambda} + \frac{(\mathbf{\Lambda}t)^2}{2!} + \frac{(\mathbf{\Lambda}t)^3}{3!} + \dots \right) \mathbf{X}^{-1} = \mathbf{X}e^{\mathbf{\Lambda}t}\mathbf{X}^{-1} \end{aligned}$$

- What is $e^{\mathbf{\Lambda}t}$?

- Example : exponential of $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \rightarrow e^{At} =$$

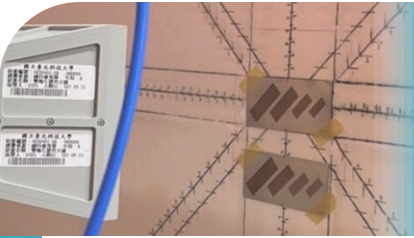
- If $\mathbf{x}_1$ is the eigenvector of $\mathbf{A}$ corresponding to eigenvalue λ_1 , then $\mathbf{x}_1$ is the eigenvector of e^{At} corresponding to eigenvalue $e^{\lambda_1 t} \rightarrow e^{At} \mathbf{x}_1 = e^{\lambda_1 t} \mathbf{x}_1$
- Matrix e^{At} is never singular. The eigenvalue $e^{\lambda t}$ is never zero
- Inverse of e^{At} always exist: e^{-At}



Differential Equation

- e^{At} regards the solution of $\frac{du}{dt} = \mathbf{A}u \rightarrow u(t) = e^{At}u_0$
- The difference equation ($u_{k+1} = \mathbf{A}u_k$) is a discrete system; however, the differential equation is a continuous system
- The solution approaches 0 if and only if all eigenvalues have a negative real part
- Like solving the difference equation, the eigenvalue fully explains the stability of the system:

Stability	Characteristic	Eigenvalue
Stable	$e^{At} \rightarrow 0$	All $\text{Re}(\lambda_i) < 0$
Neutrally stable	e^{At} is undamped	Some $\text{Re}(\lambda_i) = 0$ and other $\text{Re}(\lambda_i) < 0$
Unstable	e^{At} is unbounded	At least one $\text{Re}(\lambda_i) > 0$



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1 Introduction

2 Layer 2: Diagonalization

3 Road to Spectral Theorem

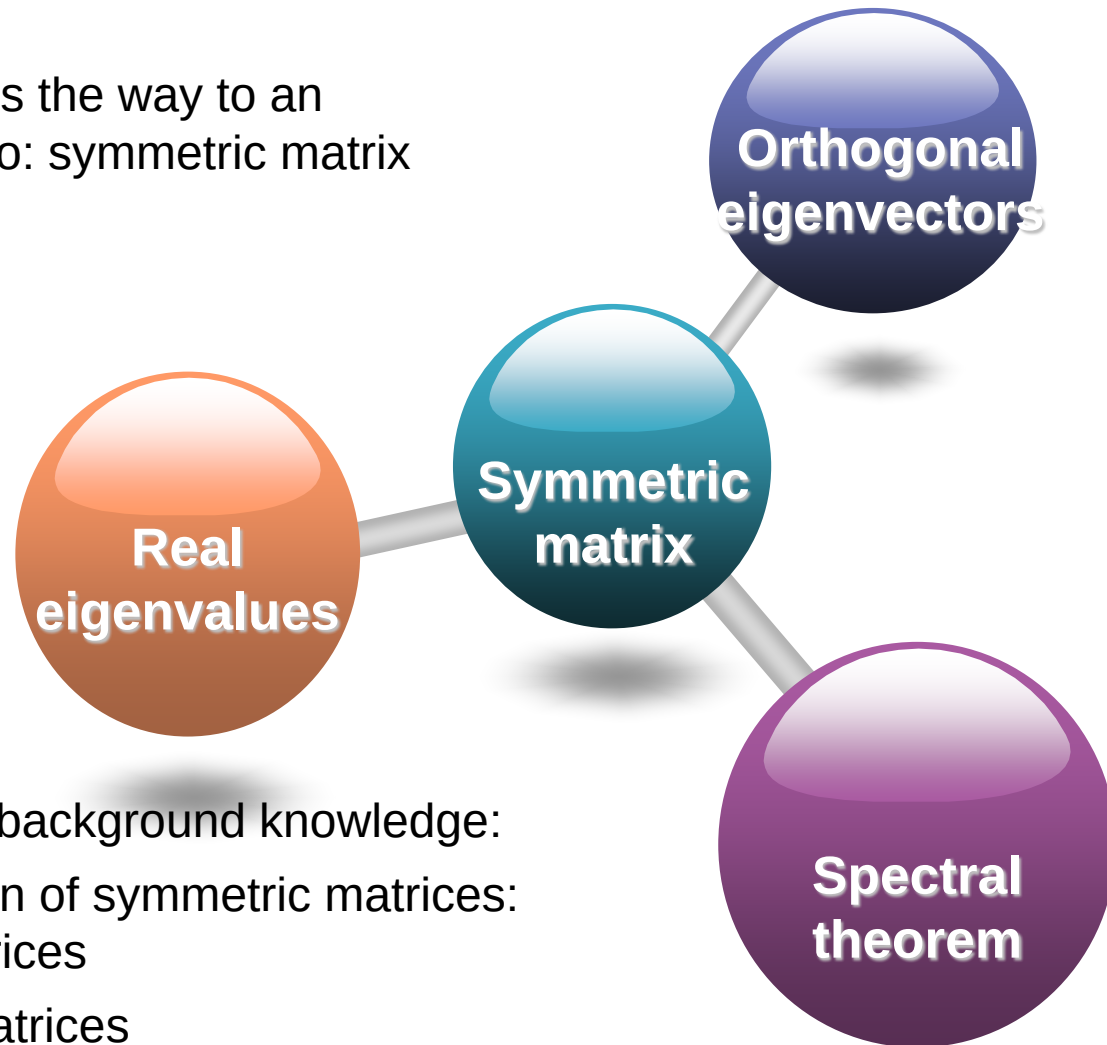
4 Layer 3: Spectral Theorem

5 Layer 1: Jordan Form

6 Layer 4: Positive Definite Matrices

Why Do We Concern Symmetric Matrices

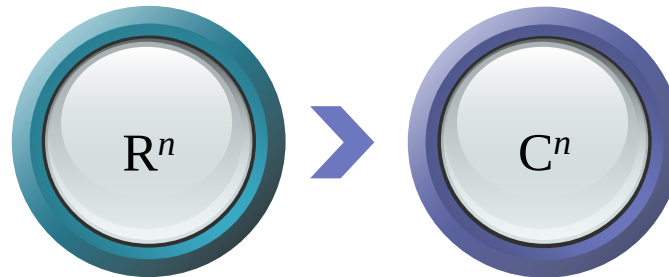
This section paves the way to an important scenario: symmetric matrix



But we need two background knowledge:

- General version of symmetric matrices: Hermitian matrices
- Similarity of matrices

Complex Numbers and Conjugates



- Three properties for complex numbers:

$$1. \overline{(a + jb)(c + jd)} = \overline{(a + jb)} \overline{(c + jd)}$$

$$2. \overline{(a + jb) + (c + jd)} = \overline{(a + jb)} + \overline{(c + jd)}$$

$$3. (a + jb)(a - jb) = a^2 + b^2 = |a + jb|^2$$

- Space C^n : all vectors $\mathbf{x}$ with n complex entries

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad x_k = a_k + jb_k$$

Comparison between Real and Complex Vectors

Characteristic	Real	Complex
Norm	$\ \mathbf{v}\ = \sqrt{ v_1 ^2 + \dots + v_n ^2} = \sqrt{\mathbf{v}^T \mathbf{v}}$	$\ \mathbf{v}\ = \sqrt{ v_1 ^2 + \dots + v_n ^2} = \sqrt{\mathbf{v}^H \mathbf{v}}$
Inner product	$\mathbf{u}^T \mathbf{v} = u_1 v_1 + \dots + u_n v_n$	$\mathbf{u}^H \mathbf{v} = \bar{u}_1 v_1 + \dots + \bar{u}_n v_n$

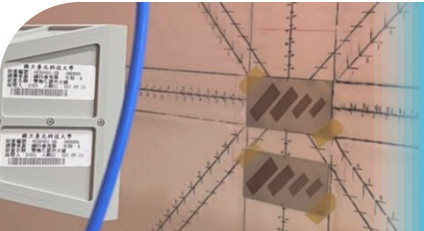
- The transpose is always accompanied with conjugate
- New operator: Hermitian transpose

vector: $\mathbf{x}^H = \overline{\mathbf{x}}^T$

matrix: $\mathbf{A}^H = \overline{\mathbf{A}}^T$

$$\begin{bmatrix} 2 + j4 & j3 \\ 7 - j3 & 5 \\ 0 & j \end{bmatrix}^H =$$

- Orthogonality of two complex vectors: $\mathbf{u}^H \mathbf{v} = 0 + j0$



Special Complex Matrices

Hermitian matrix:

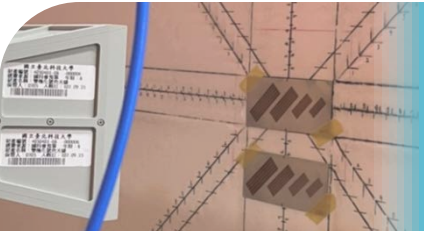
- Symmetric complex matrix
- A real symmetric matrix is a special case of Hermitian matrices
- The diagonal must be purely real

$$\mathbf{A} = \begin{bmatrix} 2 & 3 - j3 \\ 3 + j3 & 5 \end{bmatrix} = \mathbf{A}^H$$

Skew-Hermitian matrix:

- $\mathbf{K} = -\mathbf{K}^H$
- $\mathbf{K} = j\mathbf{A}$ (if $\mathbf{A}$ is Hermitian)
- The diagonal must be purely imaginary

$$\mathbf{K} = \begin{bmatrix} j2 & 3 + j3 \\ -3 + j3 & j5 \end{bmatrix} = -\mathbf{K}^H$$



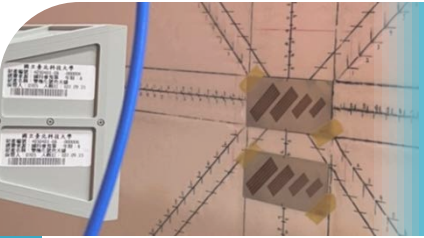
Special Complex Matrices

Unitary matrix:

- Orthogonal complex matrix
- $\mathbf{U}^H \mathbf{U} = \mathbf{I}$
- $\mathbf{U}$ preserves lengths
- Every eigenvalue of $\mathbf{U}$ has absolute value $|\lambda| = 1$
- Eigenvectors from different eigenvalues are orthogonal

$$\mathbf{U} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$$

- The three special matrices have a common feature, which will be formally introduced in Ch. 5.4



Properties of Hermitian Matrices

Property 1

- If $\mathbf{A} = \mathbf{A}^H$, $\mathbf{x}^H \mathbf{A} \mathbf{x}$ is a real number
- Proof: Find $(\mathbf{x}^H \mathbf{A} \mathbf{x})^H$

Example:

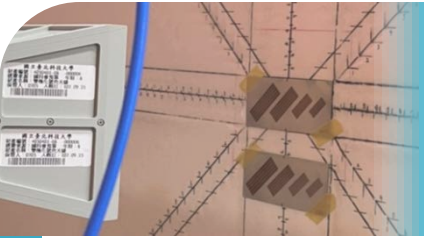
$$\begin{bmatrix} 2 & 3 - j3 \\ 3 + j3 & 5 \end{bmatrix}$$

Property 2

- Every eigenvalue of Hermitian matrices is real

Property 3

- The eigenvectors, from different eigenvalues, of a Hermitian matrix are orthogonal to one another



Properties of Skew-Hermitian Matrices

Property 1

- If $\mathbf{K} = -\mathbf{K}^H$, $\mathbf{x}^H \mathbf{K} \mathbf{x}$ is purely imaginary
- Proof: Find $(\mathbf{x}^H \mathbf{K} \mathbf{x})^H$

Example:

$$\begin{bmatrix} j2 & 3 + j3 \\ -3 + j3 & j5 \end{bmatrix}$$

Property 2

- Every eigenvalue of skew-Hermitian matrices is purely imaginary

Property 3

- The eigenvectors, from different eigenvalues, of a skew-Hermitian matrix are orthogonal to one another

Diagonalization of Hermitian Matrices

For the real-number case:

$$\text{General} \\ \mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$$



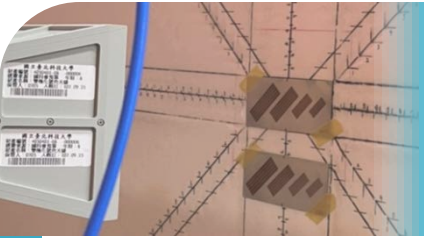
$$\text{Symmetric} \\ \mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$$

- Orthogonal eigenvectors can be chosen to be orthonormal
- $\mathbf{X}$ can be chosen to be $\mathbf{Q}$

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_n^T \end{bmatrix}$$

$$= \lambda_1 \mathbf{x}_1 \mathbf{x}_1^T + \lambda_2 \mathbf{x}_2 \mathbf{x}_2^T + \cdots + \lambda_n \mathbf{x}_n \mathbf{x}_n^T = \sum_{i=1}^n \lambda_i \mathbf{x}_i \mathbf{x}_i^T$$

$$\text{Example: } \mathbf{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$



Diagonalization of Hermitian Matrices

For the complex-number case:

Symmetric
 $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$



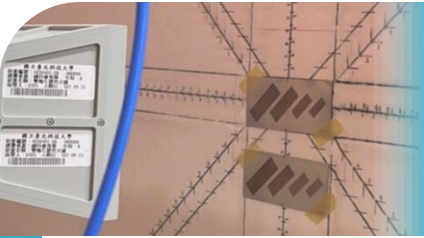
Hermitian

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H$$

Skew-Hermitian

$$\mathbf{K} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H$$

Example: $\mathbf{A} = \begin{bmatrix} 2 & 1-j \\ 1+j & 3 \end{bmatrix}$



Decomposition of a Matrix

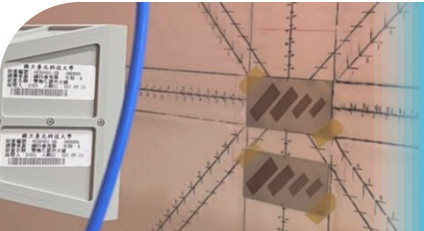
- Every matrix $\mathbf{Z}$ can be split into a Hermitian and a skew-Hermitian part
- That is, $\mathbf{Z} = \mathbf{A} + \mathbf{K}$, where $\mathbf{A} = (\mathbf{Z} + \mathbf{Z}^H)/2$ and $\mathbf{K} = (\mathbf{Z} - \mathbf{Z}^H)/2$
- Example:

$$\mathbf{Z} = \begin{bmatrix} 3 & 2 \\ 4 & 4 \end{bmatrix} \quad \mathbf{Z} = \begin{bmatrix} 3+j & 4+j2 \\ 0 & 5 \end{bmatrix}$$

- Eigenvalue of a Hermitian matrix: real, which represents decay
- Eigenvalue of a skew-Hermitian: purely imaginary, which represents oscillation
- Both $\mathbf{A}$ and $\mathbf{K}$ can be chosen to have orthonormal eigenvectors

Similarity





Similar Matrices

Definition:

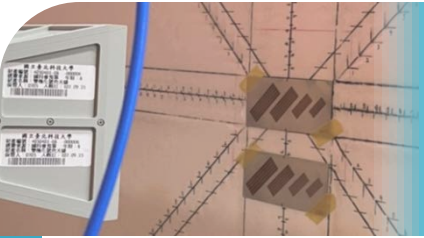
- If two matrices **A** and **B** are similar, $\mathbf{A} = \mathbf{M}\mathbf{B}\mathbf{M}^{-1}$
- In other words, **A** and **B** share the same eigenvalues

Property:

- An eigenvector **x** of **A** corresponds to an eigenvector $\mathbf{M}^{-1}\mathbf{x}$ of **B**
- Proof: Begins by $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$
- From the determinants: $\det(\mathbf{B} - \lambda\mathbf{I}) = \det(\mathbf{A} - \lambda\mathbf{I}) = 0$

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \mathbf{M} = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$



Similarity Transformation

In Ch. 2.5:

- Every linear transformation is represented by a matrix
- The transformation matrix depends on the choice of basis

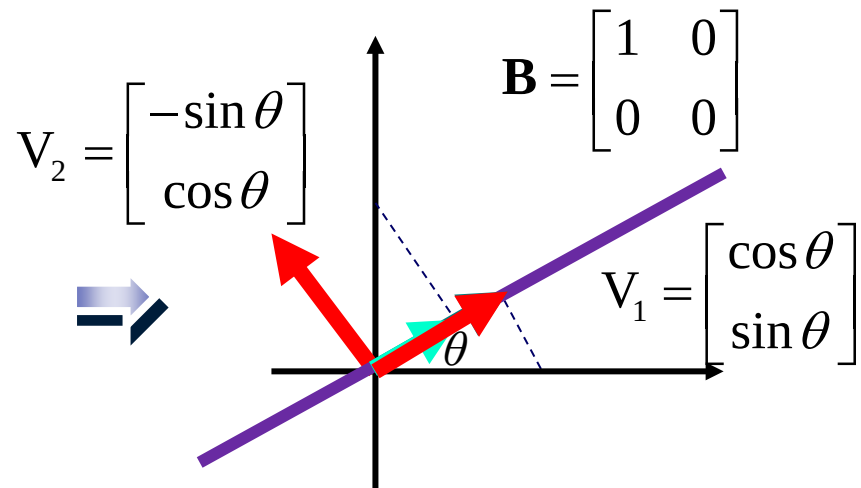
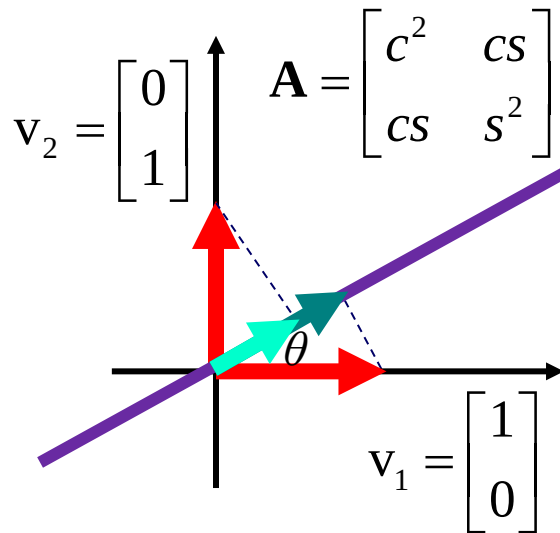
In this section:

- Similar matrices represent the same transformation with respect to different bases

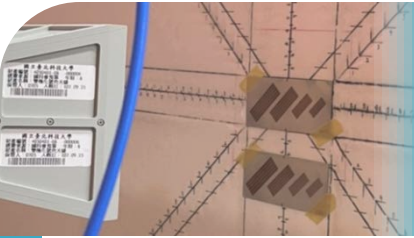
$$\begin{array}{ccccccc} & \mathbf{A} & = & \mathbf{M} & \mathbf{B} & \mathbf{M}^{-1} & \\ & \swarrow & & \swarrow & \downarrow & \searrow & \\ [\mathbf{T}]_{V \text{ to } V} & [\mathbf{I}]_{V \text{ to } V} & & [\mathbf{T}]_{V \text{ to } V} & [\mathbf{I}]_{V \text{ to } V} & & \end{array}$$

- $\mathbf{A}$ and $\mathbf{B}$ represent the same linear transformation, but different bases
- Transformation that changes only the basis is called “Identity transformation”

Example: Projection



- How do we determine $\mathbf{M}$ and $\mathbf{M}^{-1}$ efficiently?
- Elimination preserves the nullspace and row space; similarity transformation preserves the eigenvalues
- Numerically, eigenvalues are computed by a sequence of similarity transformation. The matrix goes gradually toward a triangular form, and the eigenvalues gradually appear on the diagonal



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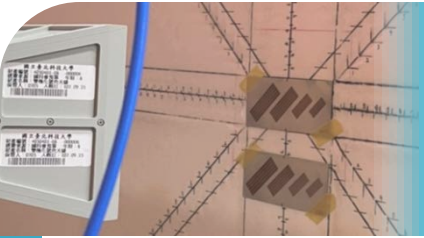
2 Layer 2: Diagonalization

3 Road to Spectral Theorem

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5 Layer 1: Jordan Form

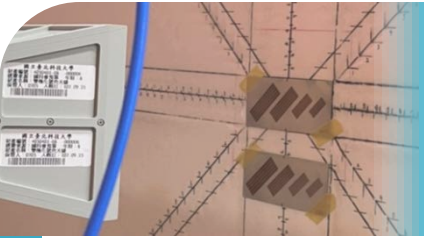
6 Layer 4: Positive Definite Matrices



Schur's Lemma

- For any matrix $\mathbf{A}$, there is a unitary matrix $\mathbf{M} = \mathbf{U}$ such that $\mathbf{U}^{-1}\mathbf{A}\mathbf{U} = \mathbf{T}$, where $\mathbf{T}$ is upper triangular
 - The eigenvalues of $\mathbf{A}$, shared by the similar matrix $\mathbf{T}$, appear along the diagonal
-
- Schur's lemma can be used to check the stability without diagonalizing $\mathbf{A}$:
 1. The power $\mathbf{A}^k$ approaches zero when all $|\lambda_i| < 1$
 2. The exponentials $e^{\mathbf{A}t}$ approaches zero when all $\text{Re}(\lambda_i) < 0$
 - Example:

$$\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} \rightarrow \mathbf{U}^{-1}\mathbf{A}\mathbf{U} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$



Spectral Theorem

Spectral theorem

- Every real symmetric matrix can be diagonalized by an orthogonal matrix
$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$$
- Every Hermitian matrix can be diagonalized by a unitary matrix
$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H$$
- The columns of $\mathbf{Q}$ or $\mathbf{U}$ contain a complete set of orthonormal eigenvectors
- Every Hermitian matrix with k different eigenvalues ($k < n$) has its own “spectral decomposition” into $\mathbf{A} = \lambda_1\mathbf{P}_1 + \lambda_2\mathbf{P}_2 + \dots + \lambda_n\mathbf{P}_n$, where $\mathbf{P}_i$ is the projection onto the eigenspace for λ_i

Example

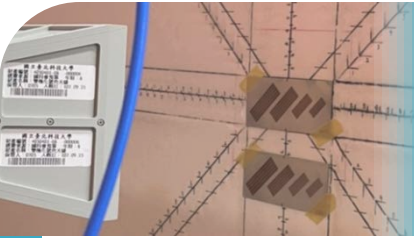
$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Repeated eigenvalues
- What are the eigenvectors?
- $\mathbf{A}$ can be decomposed by

$$\mathbf{A} = \sum_i \lambda_i \mathbf{x}_i \mathbf{x}_i^T = \lambda_1 \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \lambda_3 \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Example

$$\mathbf{A} = \lambda_1 \mathbf{P}_1 + \lambda_3 \mathbf{P}_3 = (+1) \underbrace{\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\substack{\text{Projection onto the} \\ \text{plane spanned by} \\ x_1 \text{ and } x_2}} + (-1) \underbrace{\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\substack{\text{Projection onto the} \\ \text{line with the same} \\ \text{direction as } x_3}}$$



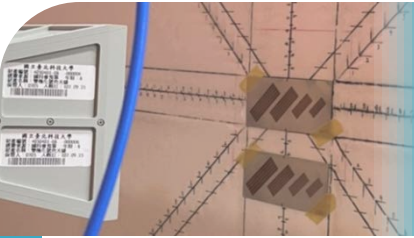
General Cases Having the Spectral Theorem

- Spectral theorem is applied to symmetric matrices, or, Hermitian matrices
- Is there any other matrix that shows $\mathbf{A} = \lambda_1 \mathbf{P}_1 + \lambda_2 \mathbf{P}_2 + \dots + \lambda_n \mathbf{P}_n$?



Normal matrices:

- Definition: $\mathbf{A}\mathbf{A}^H = \mathbf{A}^H\mathbf{A}$
- Normal matrices always have a complete set of orthonormal eigenvectors
- Example: Hermitian matrices, skew-Hermitian matrices, and unitary matrices
- If $\mathbf{A}$ is normal, $\mathbf{A}\mathbf{A}^H$ is a Hermitian matrix



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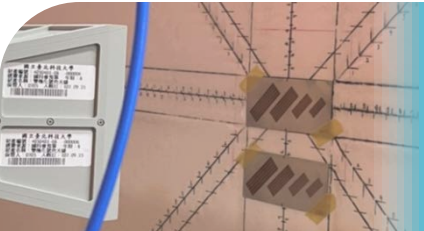
Layer 3: Spectral Theorem

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Layer 1: Jordan Form

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Layer 4: Positive Definite Matrices



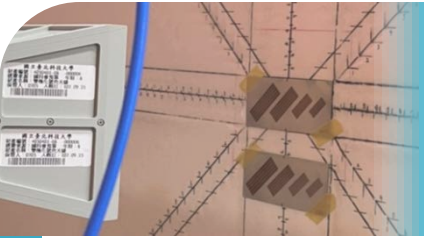
What if a Matrix Cannot be Diagonalized?

- Recall the goal of this chapter: Find a complete set of independent eigenvectors
- This enables $\mathbf{A}$ to be expressed as $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$
- A particular important case: $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ or $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H$
- When $\mathbf{A}$ cannot be diagonalized, is there anything we can do?

Jordan form:

- Goal: make $\mathbf{M}^{-1}\mathbf{A}\mathbf{M}$ as nearly diagonal as possible
- Structure: If $\mathbf{A}$ has s independent eigenvectors ($s < n$), it is similar to a matrix with s blocks

$$\mathbf{J} = \mathbf{M}^{-1}\mathbf{A}\mathbf{M} = \begin{bmatrix} \mathbf{J}_1 & & & \\ & \mathbf{J}_2 & & \\ & & \ddots & \\ & & & \mathbf{J}_s \end{bmatrix}$$



Jordan Form

- Each Jordan block $\mathbf{J}_i$ is a triangular matrix with only a single eigenvalue λ_i and only one eigenvector
- The structure of the block:

- ◆ 1×1 (no repeated eigenvalues): $\mathbf{J}_1 = [\lambda_i]$

- ◆ 2×2 (two repeated eigenvalues): $\mathbf{J}_2 = \begin{bmatrix} \lambda_i & 1 \\ 0 & \lambda_i \end{bmatrix}$

- ◆ $n \times n$ (n repeated eigenvalues): $\mathbf{J}_n = \begin{bmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix}$

- When the block has order $m > 1$, the eigenvalue λ_i repeats m times, and there are $m - 1$ 1's above the diagonal
- Two matrices shared the same Jordan form are similar

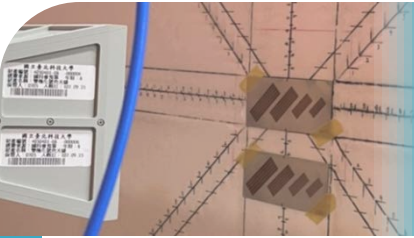
Examples of Jordan Form

$$\mathbf{A} = \begin{bmatrix} 5 & 4 & 2 & 1 \\ 0 & 1 & -1 & -1 \\ -1 & -1 & 3 & 0 \\ 1 & 1 & -1 & 2 \end{bmatrix}$$

- Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 4$, and $\lambda_4 = 4$
- Eigenvectors: $\mathbf{x}_1 = [-1 \ 1 \ 0 \ 0]^T$, $\mathbf{x}_2 = [1 \ -1 \ 0 \ 1]^T$, and $\mathbf{x}_3 = \mathbf{x}_4 = [1 \ 0 \ -1 \ 1]^T$
- Thus, $\mathbf{A}$ cannot be diagonalized
- Jordan form for $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} -1 & 1 & 1 & \color{red}{1} \\ 1 & -1 & 0 & \color{red}{0} \\ 0 & 0 & -1 & \color{red}{0} \\ 0 & 1 & 1 & \color{red}{0} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 & \color{red}{1} \\ 1 & -1 & 0 & \color{red}{0} \\ 0 & 0 & -1 & \color{red}{0} \\ 0 & 1 & 1 & \color{red}{0} \end{bmatrix}^{-1}$$

↪ "Generalized eigenvector" from $\mathbf{A}\mathbf{v}_{j+1} = \lambda\mathbf{v}_{j+1} + \mathbf{v}_j$



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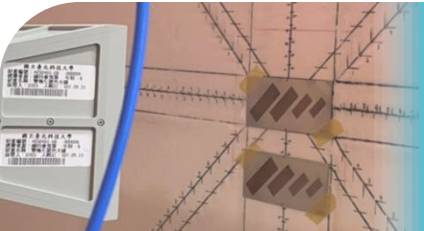
Layer 3: Spectral Theorem

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Layer 1: Jordan Form

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Layer 4: Positive Definite Matrices



Roadmap in This Chapter

Arbitrary matrix



Diagonalizable matrix



Symmetric matrix



Positive definite symmetric matrix

- $\mathbf{Ax} = \lambda \mathbf{x}$
- But some matrices fail to provide n independent eigenvectors
- $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$
- Now any new vector can be expressed as the linear combination of eigenvectors
- $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$
- Full set of eigenvectors can be found even for repeated λ
- Eigenvalues are always real
- Eigenvectors are orthonormal
- Spectral theorem
- Nonsingular $\rightarrow$ invertible
- Energy $\mathbf{x}^T \mathbf{Ax} > 0$
- All pivots > 0
- All leading determinants > 0
- All eigenvalues > 0
- $\mathbf{A} = \mathbf{R}^T \mathbf{R}$

Quadratic Function in Matrix Form

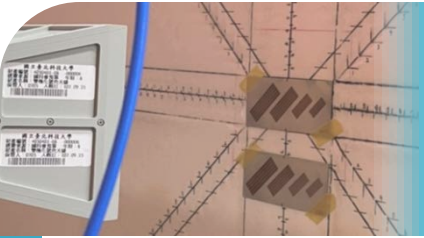
$$f = ax^2 + 2bxy + cy^2 = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{x}^T \mathbf{A} \mathbf{x}$$

- General case:

$$f = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j = a_{11}x_1^2 + a_{12}x_1x_2 + a_{12}x_2x_1 + \cdots + a_{nn}x_n^2$$
$$= \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{x}^T \mathbf{A} \mathbf{x}$$

- Example:

$$f = 2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_3^2 \Rightarrow \mathbf{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$



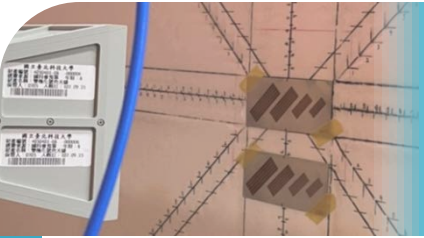
Characteristics of $\mathbf{x}^T \mathbf{A} \mathbf{x}$

- $\mathbf{A}$ is a symmetric matrix
- f has a minimum when $f = \mathbf{x}^T \mathbf{A} \mathbf{x}$ is positive
- Considering only $f = ax^2 + 2bxy + cy^2$
 - ◆ Stationary point: the point at which $\partial f / \partial x = 0$ and $\partial f / \partial y = 0$

$$f = ax^2 + 2bxy + cy^2 = a \left(x + \frac{b}{a} y \right)^2 + \left(c - \frac{b^2}{a} \right) y^2$$

- ◆ f is strictly positive if $a > 0$ and $ac > b^2$
- ◆ Algebraically: Positive definite
- ◆ Geometrically: a bowl

- ◆ f is strictly negative if $a < 0$ and $ac > b^2$
- ◆ Algebraically: Negative definite
- ◆ Geometrically: a upside down bowl



Characteristics of $\mathbf{x}^T \mathbf{A} \mathbf{x}$

- Other possibilities:
 - ◆ f is nonnegative if $a \geq 0$ and $ac \geq b^2$
 - ◆ Algebraically: Positive semidefinite
 - ◆ Geometrically: a trough, degenerated from a bowl

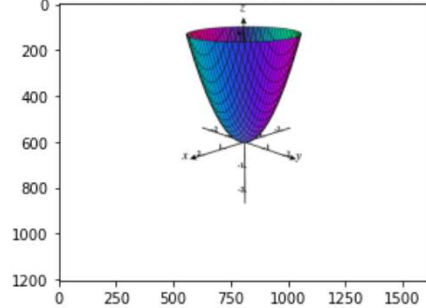
 - ◆ f is strictly negative if $a \leq 0$ and $ac \geq b^2$
 - ◆ Algebraically: Negative semidefinite
 - ◆ Geometrically: a upside down trough

 - ◆ f is ambiguous if $ac < b^2$
 - ◆ Algebraically: Indefinite
 - ◆ Geometrically: a saddle

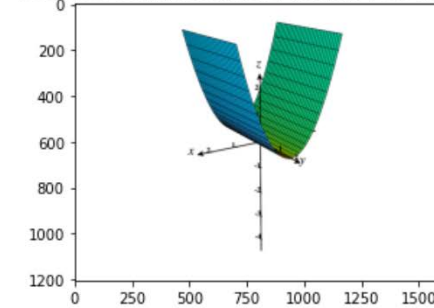
*Such information
is revealed by Δ !*

Geometry

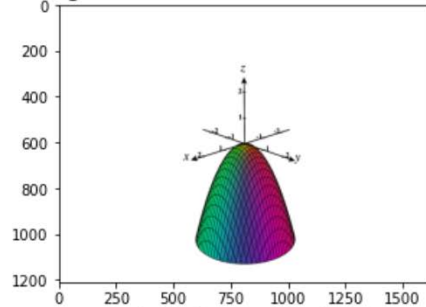
Positive Definite: Absolute Minima



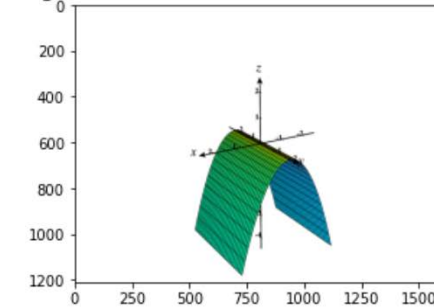
Positive Semi-Definite: Minima on an Axis



Negative Definite: Absolute Maxima



Negative Semi-Definite: Maxima on an Axis



Indefinite: Saddle Point

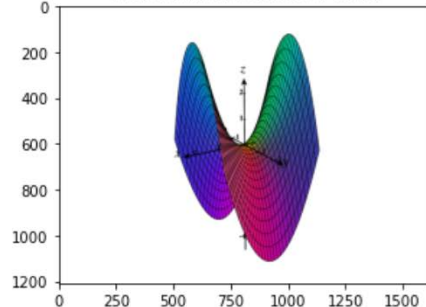
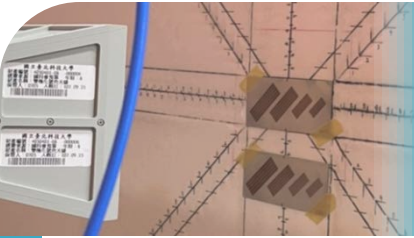


Image Credit: Ximera, OSU - Categorizing Quadratic Forms



Positive Definite Quadratic Function

$$f = ax^2 + 2bxy + cy^2 = a\left(x + \frac{b}{a}y\right)^2 + \left(c - \frac{b^2}{a}\right)y^2 = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{x}^T \mathbf{A} \mathbf{x}$$

- To have $a > 0$ and $ac > b^2$:
 1. Both eigenvalues are positive
 2. Leading determinants are positive
 3. Both pivots are positive

Tests for a symmetric matrix $\mathbf{A}$ to be positive definite

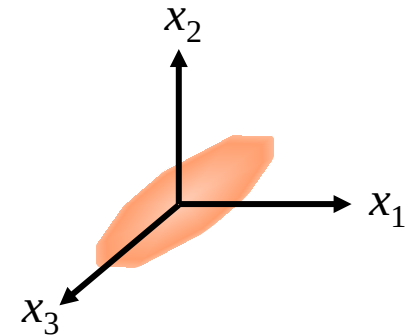
- Energy $\mathbf{x}^T \mathbf{A} \mathbf{x} > 0$ for all nonzero vectors (definition)
- All the eigenvalues of $\mathbf{A}$: $\lambda_i > 0$
- All the upper left submatrices $\mathbf{A}_k$ have positive determinants
- All the pivots of $\mathbf{A}$: $d_i > 0$
- There is a matrix $\mathbf{R}$ with independent columns generating $\mathbf{A} = \mathbf{R}^T \mathbf{R}$

Ellipsoids in n Dimensions

- Function of an ellipse or ellipsoid: $\mathbf{x}^T \mathbf{A} \mathbf{x} = 1$

- Example:

$$\mathbf{A} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{9} \end{bmatrix} \rightarrow \mathbf{x}^T \mathbf{A} \mathbf{x} = 4x_1^2 + x_2^2 + \frac{1}{9}x_3^2 = 1$$

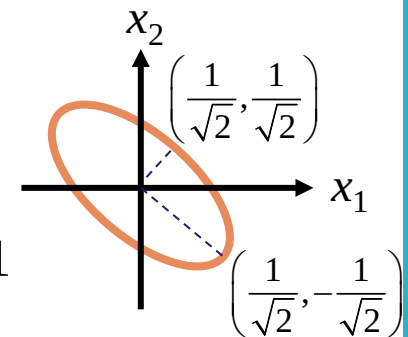


An ellipsoid without rotation

- Another example:

$$\mathbf{A} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \rightarrow \mathbf{x}^T \mathbf{A} \mathbf{x} = 5x_1^2 + 8x_1x_2 + 5x_2^2 = 1$$

$$5x_1^2 + 8x_1x_2 + 5x_2^2 = \left(\frac{x_1}{\sqrt{2}} - \frac{x_2}{\sqrt{2}} \right)^2 + 9 \left(\frac{x_1}{\sqrt{2}} + \frac{x_2}{\sqrt{2}} \right)^2 = 1$$



An ellipse rotated by 45°

- What do the eigenvalues and eigenvectors mean in this ellipse?

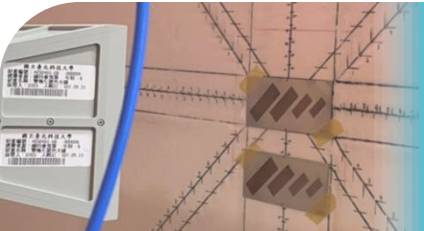
Ellipsoids and Eigenvalues/Eigenvectors

Matrix	Eigenvector	Eigenvalue
	$\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}^T$	1
	$\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}^T$	9

Ellipsoid	Direction	Length
Major axis	$\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}^T$	1
Minor axis	$\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}^T$	$1/\sqrt{9} = 1/3$

$$\mathbf{x}^T \mathbf{A} \mathbf{x} = (\mathbf{x}^T \mathbf{Q}) \mathbf{\Lambda} (\mathbf{Q}^T \mathbf{x}) = \mathbf{y}^T \mathbf{\Lambda} \mathbf{y} = \lambda_1 y_1^2 + \dots + \lambda_n y_n^2$$

- $\mathbf{x} \rightarrow \mathbf{y} = \mathbf{Q}^T \mathbf{x}$: Rotate the axes to match the axes of the ellipsoid
- $\mathbf{x}^T \mathbf{A} \mathbf{x} = 5x_1^2 + 8x_1x_2 + 5x_2^2 = 1 \rightarrow y_1^2 + 9y_2^2 = 1$



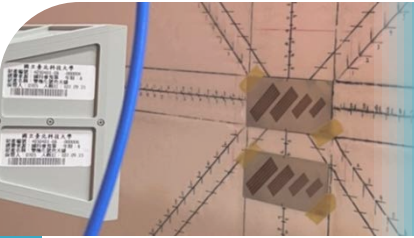
Ellipsoids and Eigenvalues/Eigenvectors

- Suppose $\mathbf{A}$ is positive definite and symmetric: $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ with $\lambda_i > 0$
- The rotation $\mathbf{y} = \mathbf{Q}^T\mathbf{x}$ simplifies $\mathbf{x}^T\mathbf{A}\mathbf{x} = 1$ to

$$(\mathbf{x}^T\mathbf{Q})\mathbf{\Lambda}(\mathbf{Q}^T\mathbf{x}) = 1 \quad \Rightarrow \quad \mathbf{y}^T\mathbf{\Lambda}\mathbf{y} = 1 \quad \Rightarrow \quad \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 = 1$$

Equation of an ellipsoid

- The direction of the axes: eigenvectors (in the original $\mathbf{x}$ space)
- The lengths of the axes: $\frac{1}{\sqrt{\lambda_1}}, \frac{1}{\sqrt{\lambda_2}}, \dots, \frac{1}{\sqrt{\lambda_n}}$



Semidefinite Matrices

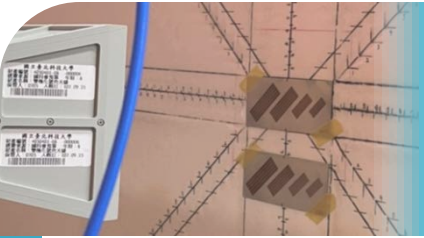
Tests for a symmetric matrix $\mathbf{A}$ to be positive semidefinite

- Energy $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$ for all nonzero vectors (definition)
- All the eigenvalues of $\mathbf{A}$: $\lambda_i \geq 0$
- All the upper left submatrices $\mathbf{A}_k$ have nonnegative determinants
- All the pivots of $\mathbf{A}$: $d_i \geq 0$
- There is a matrix $\mathbf{R}$, possibly with dependent columns, generating $\mathbf{A} = \mathbf{R}^T \mathbf{R}$

- Examples:

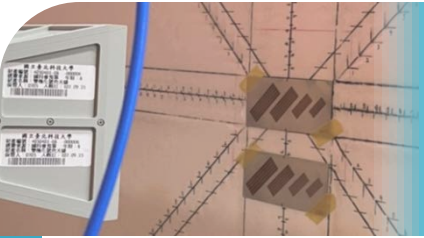
$$\mathbf{A} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$



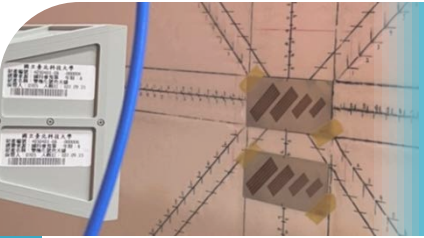
ConceptTests

- Why are we interested in finding a full set of independent eigenvectors?
 - 1) To understand the stability and behavior of dynamical systems
 - 2) To find the Jordan form of a matrix
 - 3) To make these eigenvectors serve the space as a basis
 - 4) To test whether the linear transformation can be inverted



ConceptTests

- When we find eigenvalues and eigenvectors for a symmetric matrix, which of the following statements are true?
 - 1) The eigenvalues are always real
 - 2) A symmetric matrix must have distinct eigenvalues
 - 3) Symmetric matrices can always be diagonalized
 - 4) All symmetric matrices are invertible
 - 5) The eigenvectors are always orthogonal



ConceptTests

- What are the characteristics about a positive definite symmetric matrix $\mathbf{A}$?
 - 1) For any non-zero vector $\mathbf{x}$, $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$
 - 2) All its eigenvalues are positive
 - 3) All its pivots are positive
 - 4) $\mathbf{A}$ is always invertible
 - 5) All its eigenvectors are orthogonal